

PRAMĀNA

प्रमाण · *proof*

Every state already knows how bad its air is.
None can prove whose it is.

Source-receptor attribution for cross-border air pollution — computed from live satellite, weather and CPCB data, sealed in a certificate the accused state can re-run.



Sense



Attribute



Seal

Every winter, the same argument nobody can settle

317

Delhi receptor AQI

Worst live CPCB station (DTU), read by the system on 15 Sep 2026

41

Delhi CPCB stations

reporting in that same pull — the data exists

228 km

Punjab → Delhi corridor

smoke crosses two state lines before it arrives

- Delhi blames upwind crop burning. Punjab and Haryana point at Delhi's own traffic and construction.
- Dashboards show how bad the air is — not whose it is, and not with a number another state can check.
- Advisories go out in English on a portal, not in the languages people along the corridor speak.

Measuring the wrong layer, and publishing numbers nobody can check

1

The wind that carries smoke is not the wind stations measure

Crop smoke rides the 925 hPa layer, about 750 m up. At night a temperature inversion decouples it from the ground, so the 10 m wind a monitoring station reports can point somewhere the smoke is not going.

2

A source share nobody can reproduce is just an opinion

Studies disagree, methods are unpublished, and the state being named has no way to re-run the finding. So every attribution becomes a negotiating position.

3

Warnings that do not reach people are not warnings

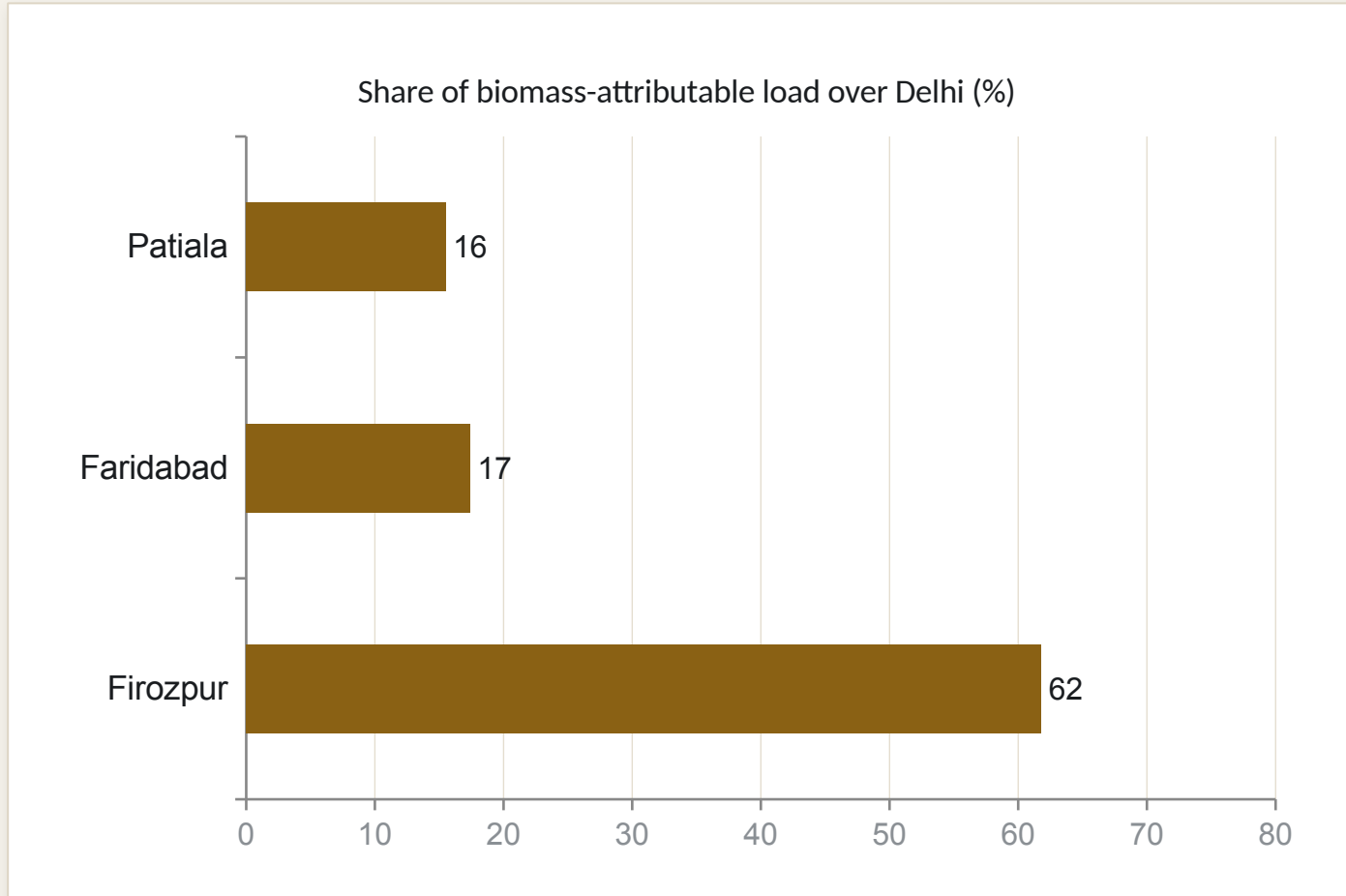
Outdoor workers along the corridor read Punjabi, Hindi, Urdu — or do not read at all. An English web notice misses the people most exposed.

One pipeline: from a fire in Punjab to a sealed finding and a spoken warning



Live at /console, /report, /advisory and /certificate — the same run feeds all four, so the advisory, the notice and the certificate cannot disagree.

A register of upwind tehsils, with confidence bands



Computed, not recalled

Leading sources from one engine run on 15 Sep 2026 inputs, on the recorded 3 Nov wind field.



Honest about its limits

A satellite sees crop fires, not traffic or industry. These are shares of the biomass load, and the page says so.



Live mode says when nothing arrives

On today's September wind, little of Punjab's burning reaches Delhi — and the console reports exactly that.

Gemini where a model helps — never where proof must



Gemini multimodal vision

Reads a citizen's sky photo into a bounded AQI band, visibility and likely source — with caveats. Refuses screenshots and graphics outright. One lone report is capped at 0.35 of the model's confidence until neighbours corroborate it.



Gemini structured reasoning

Drafts the operator advisory and the notice to the named state, as schema-constrained JSON from the computed register — not free text.



Built to survive a free tier

Calls fall through gemini-3.6 → 3.7 → 3.5 → 3.8-flash on quota or overload, and record which model answered.

Deliberately no model here

Attribution

Physics, not a learned model — a state being named must be able to re-run it offline and get the same answer.

Sealing

SHA-256 and a Merkle tree. A certificate that depended on a hosted model could not be reproduced.

A certificate the accused state can re-run

FIRMS active fire detections

sha256 of the exact data consumed

925 hPa wind field

sha256 of the exact data consumed

CPCB receptor network

sha256 of the exact data consumed

Attribution register

sha256 of the exact data consumed

Model configuration

sha256 of the exact data consumed



Merkle root

```
6e7dbed47171597de5faf46b952830f3
2e104222545926c088dd0b2d1527ccbc
```

- Recomputed in the browser: root reproduced, all 5 inputs and every internal node match.
- Same inputs → same id and root. Change one input and the leaf that moved says which.
- Certificate PRM-CERT-20260915-E-F938C03B9E is carried on both the advisory and the notice.

Four users, one chain of evidence

1

Air quality commissions and state pollution boards

A contribution register by tehsil, and a cross-state notice that leads with a verifiable certificate instead of an accusation.

2

District magistrates

An advisory tied to the live CPCB reading and GRAP stage, with actions ordered by exposure averted.

3

Citizens along the corridor

Warnings in 12 Indian languages, read aloud; report the sky with a phone photo; one report cannot move the register alone.

4

Courts and tribunals

Evidence that either side can recompute, with the model's limitations printed next to its numbers.

Pilot-ready on public data and existing infrastructure



No new hardware

Runs on networks India already operates — CPCB stations, NASA FIRMS satellites, public weather models — plus phones people already carry.



Works with zero keys, goes live one key at a time

Every capability checks for its key at call time. Without one it serves recorded output and says so on screen — nothing silently fakes a result.



Two deployment forms

A live web deployment today, and a Cloud Run container that authenticates to Vertex AI as a service account — no key material in a government environment.



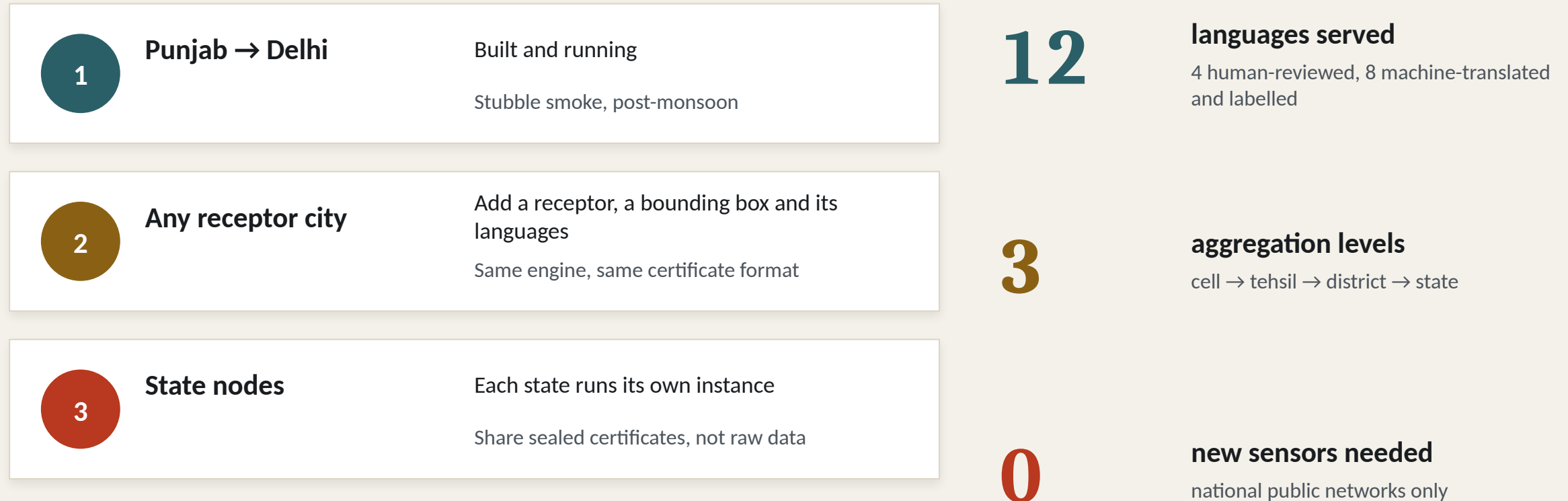
A human signs what crosses a border

Notices are drafted, never auto-sent. Dispatch is an officer's decision.

Live on the demo

● Gemini vision	live
● Gemini reasoning	live
● CPCB via data.gov.in	live
● NASA FIRMS	live
● Open-Meteo 925 hPa	live
● Translation / TTS / STT	wired
● Maps Platform	wired

A corridor is configuration, not a rebuild



The same design extends beyond India: the certificate format is what lets neighbouring jurisdictions share a finding without sharing their raw monitoring data.

From a working prototype to a pilot

1

Validate

Score the transport model against past CPCB episodes and publish where it is wrong.

2

Deepen inputs

Add TROPOMI columns via Earth Engine for industrial and urban sources.

3

Persist

Move the citizen register to Firestore and archive certificates in BigQuery.

4

Review

Have state health departments sign off the eight machine-translated languages.

Try it

Live demo

pramana-arya-patils-projects-37178a61.vercel.app

Source code

github.com/Arya-Patil686/pramana

Not a dashboard that shows pollution — an instrument that settles who owns it.